



Circular Motion

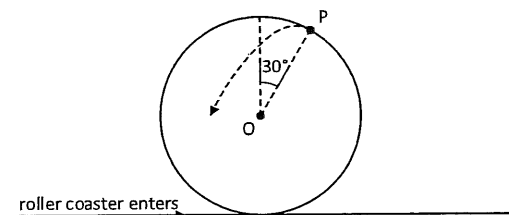
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Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs (ACJC – NJC)

1. [CJC/H2/Prelims 2018/P1/10]

A toy roller coaster attempts a loop-de-loop. It enters the bottom of loop of radius 1.0 m with insufficient speed. It loses contact at the point P as shown. The line joining P and the centre of the loop O makes an angle 30° with the vertical.



What is the speed at P which the roller coaster just loses contact with the loop?

- A 0 m s^{-1} B 2.2 m s^{-1} C 2.9 m s^{-1} D 3.1 m s^{-1}

2. [AJC/H2/Prelims 2018/P1/8]

An electric scooter of mass 10 kg moves at a constant speed over a humpback bridge of radius of curvature 3.0 m. What is maximum speed of the electric scooter such that it does not lose contact with the bridge?

- A 1.8 m s^{-1} B 3.3 m s^{-1} C 5.4 m s^{-1} D 5.7 m s^{-1}

3. [CJC/H2/Prelims 2018/P1/7]

A car is travelling on a hump with a radius of curvature of 30 m as shown in figure below. The car loses contact with the hump at the highest point.

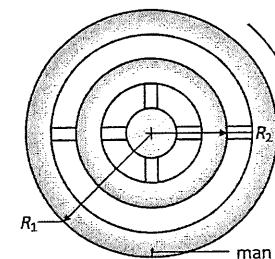


At what speed will the car be losing contact as it moves over the hump at the highest point?

- A 15.7 m s^{-1} B 17.2 m s^{-1} C 22.2 m s^{-1} D 29.4 m s^{-1}

4. [CJC/H2/Prelims 2018/P1/8]

A proposed space laboratory is to create artificial gravity as shown in the figures below. The space laboratory is rotated about an axis so that it simulates an acceleration due to gravity equal to that of the gravity due to Earth at the outer ring which has a radius R_1 of 2150 m.

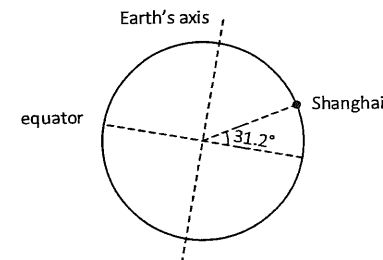


What should be the approximate radius R_2 of the inner ring, so that it simulates the acceleration due to the gravity of Mars which is 3.72 m s^{-2} ?

- A 700 m B 800 m C 900 m D 1000 m

5. [CJC/H2/Prelims 2018/P1/9]

A person is located near the city of Shanghai which has a latitude of 31.2° N as shown in the figure below.



Assuming that the Earth is a sphere of radius 6380 km, the linear velocity of the person due to the rotation of the Earth about its axis is

- A 240 m s^{-1} B 337 m s^{-1} C 397 m s^{-1} D 464 m s^{-1}

6. [DHS/H2/Prelims 2018/P1/9]

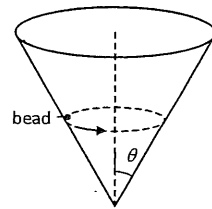
The reading of a speedometer fitted to the front wheel of a bicycle is directly proportional to the angular velocity of the wheel. A certain speedometer is correctly calibrated for use with a wheel of diameter 61 cm but, by mistake, is fitted to a 51 cm wheel.

What is the value of $\frac{\text{indicated speed} - \text{actual speed}}{\text{actual speed}} \times 100\%$

- A +16% B -16% C +20% D -20%

7. [EJC/H2/Prelims 2018/P1/7]

A small bead is set into circular motion in a horizontal plane within a smooth conical cone as shown below.



If the bead is moving with a centripetal acceleration of g , what is the angle θ ?

- A 30° B 45° C 55° D 70°

8. [HCI/H2/Prelims 2018/P1/10 Modified]

Two discs of mass m and $2m$ are placed on a rough, horizontal and level turntable as shown in the Fig. 8.1. The turntable starts rotating from rest with gradually increasing angular velocity ω about a vertical axis through its centre O. Eventually, both discs will slip off the turntable.

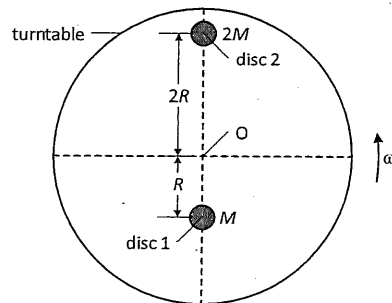


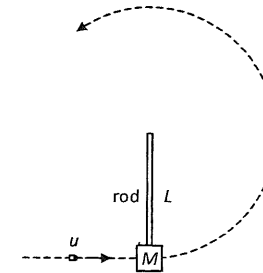
Fig. 8.1

Given that the maximum frictional force acting on disc 1 is half of that on disc 2, which of the following statements is correct?

- A Disc 1 experiences maximum frictional force first.
 B Disc 2 experiences maximum frictional force first.
 C Both discs experience maximum frictional force at the same time.
 D Neither disc will experience maximum frictional force when they slipped off the turntable.

9. [HCI/H2/Prelims 2018/P1/11 Modified]

A block of mass M is suspended by a stiff rod of length L and negligible mass. A bullet of mass m and horizontal velocity u strikes the block. The bullet remains in the block which swings upwards.

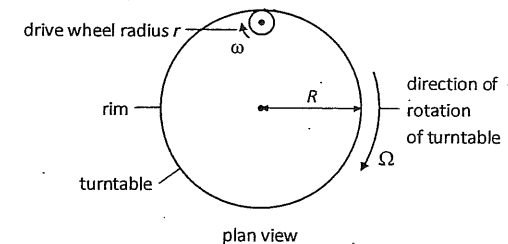


What is the minimum value of u such that the block and the bullet complete the vertical circle? Ignore air resistance.

- A $u = \frac{4(m+M)gL}{m}$ B $u = \frac{5(m+M)gL}{m}$
 C $u = \frac{2(m+M)\sqrt{gL}}{m}$ D $u = \frac{(m+M)\sqrt{5gL}}{m}$

10. [UC/H2/Prelims 2018/P1/9]

A turntable has radius R . It is driven by a rubber drive wheel of radius r in contact with the inside of the rim of the turntable, as shown in the plan view diagram.



The turntable rotates with angular velocity Ω and the linear speed of a point on its rim is V . The drive wheel rotates with angular velocity ω and the linear speed of a point on its rim is v .

Which pair of equations show the relationship between the angular velocities and the linear speeds of the turntable and the wheel?

	angular velocities	linear speeds
A	$\Omega = \omega$	$V = v$
B	$\Omega = \omega$	$V = \left(\frac{r}{R}\right)v$
C	$\Omega = \left(\frac{r}{R}\right)\omega$	$V = v$
D	$\Omega = \left(\frac{R}{r}\right)\omega$	$V = \left(\frac{r}{R}\right)v$

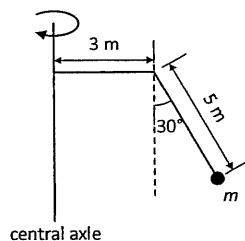
11. [JJC/H2/Prelims 2018/P1/7]

A particle travels at a constant speed around a circle of radius r with centripetal acceleration a . What is the time taken for ten complete revolutions?

- A $\frac{\pi}{5}\sqrt{\frac{a}{r}}$ B $\frac{\pi}{5}\sqrt{\frac{r}{a}}$ C $20\pi\sqrt{\frac{a}{r}}$ D $20\pi\sqrt{\frac{r}{a}}$

12. [JJC/H2/Prelims 2018/P1/8]

A mass m , attached to one end of an inextensible string of length 5.0 m, is made to move in a horizontal circle as the vertical central axle is rotated. In a particular circular motion, the string makes an angle of 30° with the vertical as shown below.



What is the period of the mass about the central axle?

- A 1.0 s B 3.5 s C 4.8 s D 6.2 s

13. [MI/H2/Prelims 2018/P1/12]

A ball is inside a pail that is moving in a vertical circle of radius 70.0 cm. The ball will just remain in contact with the pail at the top of the circular path if the pail is rotating with a minimum speed v .

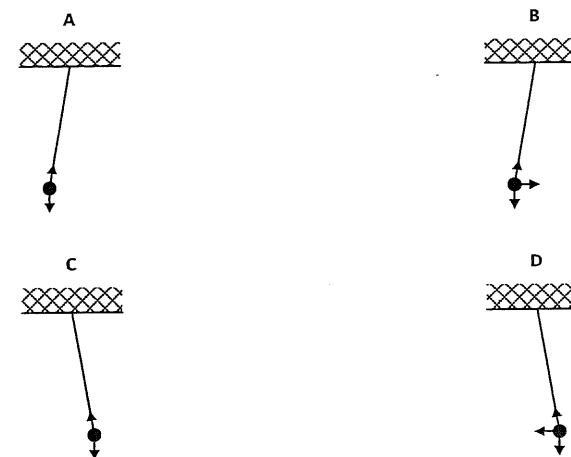
What is the minimum speed v of the pail?

- A 2.6 m s^{-1} B 3.7 m s^{-1} C 26.0 m s^{-1} D 37.0 m s^{-1}

14. [MJC/H2/Prelims 2018/P1/7]

A passenger is sitting in a railway carriage facing in the direction in which the train is travelling. A pendulum hangs down in front of him from the carriage roof. The train travels along a circular arc bending to the left.

Which one of the following diagrams shows the position of the pendulum as seen by the passenger, and the directions of the forces acting on it?



15. [NJC/H2/Prelims 2018/P1/7 Modified]

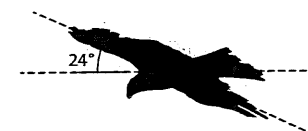
A stone is attached to one end of a light inextensible string. The other end of the string is attached to a fixed point and the stone moves in a vertical circle.

Which of the following statements is false?

- A The tension of the string is greater than the weight of the stone at the lowest point.
 B The velocity of the stone is the same as long as the stone is at the same height from the ground.
 C The magnitude of the tension increases when the stone moves from the highest position to the lowest position.
 D The difference in the tension when the stone is at the top and the bottom of the circle is proportional to the mass of the stone.

16. [NJC/H2/Prelims 2018/P1/8]

A bird is soaring in a horizontal circular path of radius 2.0 m. Its bank angle relative to the horizontal is 24° as shown in the diagram below.

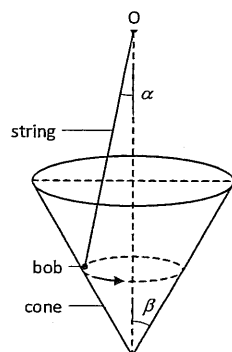


What is the speed of the bird?

- A 1.5 m s^{-1} B 3.0 m s^{-1} C 6.6 m s^{-1} D 8.7 m s^{-1}

17. [NYJC/H2/Prelims 2018/P1/10]

A pendulum bob at the end of the string which is fixed at O moves in a horizontal circle on the inside of a cone as shown.



If the tension in the string is T and the normal contact force on the bob is N , which of the following gives the centripetal force on the bob?

- A $T \sin \alpha + N \cos \beta$ B $T \sin \alpha - N \cos \beta$
 C $T \cos \alpha + N \sin \beta$ D $T \cos \alpha - N \sin \beta$

18. [PJC/H2/Prelims 2018/P1/6 Modified]

A turntable is rotating about a vertical axis through its centre. Two discs of mass 3.0 kg and 6.0 kg are placed on the rough surface of the turntable at a distance of 2.0 m and 5.0 m from its centre respectively.

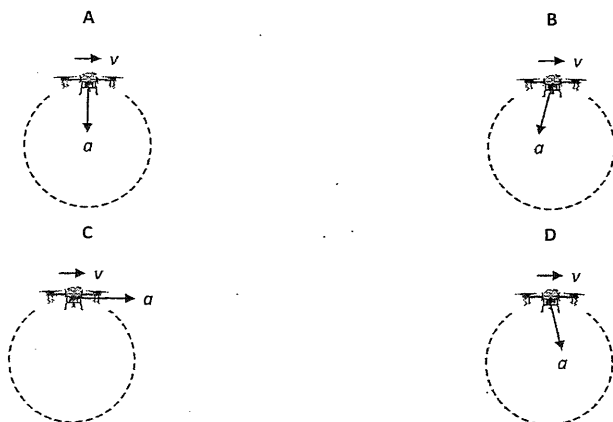
What is the ratio of the centripetal force on the 3.0 kg disc to the 6.0 kg disc?

- A 0.20 B 0.80 C 5.0 D 1.3

19. [RVHS/H2/Prelims 2018/P1/10 Modified]

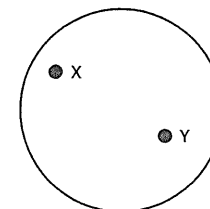
A drone is performing a stunt for its audience on the ground by flying in a vertical circle.

Which diagram shows the acceleration a of the drone at its highest point where its velocity is v and is speeding up?



20. [SRJC/H2/Prelims 2018/P1/11 Modified]

An equal amount of sand is sprinkled on a turntable at points X and Y. The turntable is rotating with increasing speed.

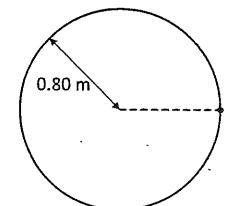


Which one of the following comparing sand at points X and Y is true?

	angular speed	linear speed	which will fly off first?
A	$X = Y$	$X > Y$	X
B	$X = Y$	$X > Y$	Y
C	$X > Y$	$X = Y$	Y
D	$X > Y$	$X > Y$	Y

21. [SRJC/H2/Prelims 2018/P1/12]

A model car moves in a circular path of radius 0.80 m at an angular speed of $0.5\pi \text{ rad s}^{-1}$.



What is its displacement from point P, 4.0 s after passing P?

- A zero B 1.6 m C $0.8\pi \text{ m}$ D $1.6\pi \text{ m}$

22. [TPJC/H2/Prelims 2018/P1/13 Modified]

A car, mass m , drives over a bridge of radius of curvature r with a constant speed v .

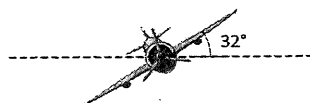


What is the force on the car by the bridge when it is at the top of the bridge?

- A mg B $\frac{mv^2}{r} + mg$
 C $\frac{mv^2}{r} - mg$ D $mg - \frac{mv^2}{r}$

23. [VJC/H2/Prelims 2018/P1/7 Modified]

A fighter plane flies with constant speed v in a horizontal circle of radius r . To do so, it banked its wings at an angle of 32° to the horizontal.

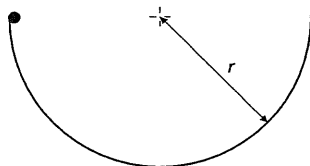


At what angle to the horizontal must it bank its wing to fly in a horizontal circle of radius $r/2$ with a speed of $2v$?

- A 17° B 51° C 68° D 79°

24. [YJC/H2/Prelims 2018/P1/7]

A small object of weight mg is released from rest at the rim of a smooth semi-spherical bowl of radius r .



What is the magnitude of the normal force, in terms of mg , acting on the object when it passes the bottom of the bowl?

- A mg B $2mg$ C $2.5mg$ D $3mg$

25. [YJC/H2/Prelims 2018/P1/8 Modified]

A car travels over a hump of radius of curvature r . At the highest point of the hump, the driver whose safety belt is not secured, experiences weightlessness.

What is the speed of the car, in terms of r and g , at the highest point of the hump?

- A $\sqrt{\frac{rg}{2}}$ B $\frac{\sqrt{rg}}{2}$ C $\sqrt{rg}$ D rg

II. Structured Questions (AJC – VJC)

1. [AJC/H2/Prelims 2018/P2/2]

A circular disc of radius 11.8 cm is spinning about its centre O in a vertical plane at a rate of 100 revolutions per minute. A plasticine of mass 3.8 g is stuck to the edge of the disc at point P and the line OP is 45° from the vertical at the instant, as shown in Fig. 1.1.

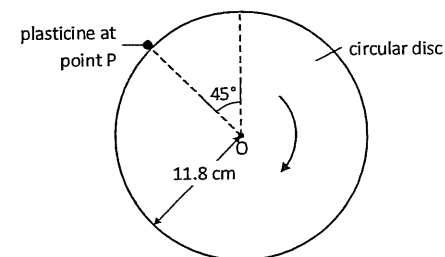


Fig. 1.1

- a. Show that the centripetal force acting on the plasticine is 0.049 N. [3]
 b. On Fig. 1.2, the weight of the plasticine at P has been drawn. At this instant, the magnitude of the contact force by the disc on the plasticine is equal to the weight of the plasticine. Draw an arrow on Fig. 1.2 to show the contact force by the disc on the plasticine at P. Label this arrow C. [1]

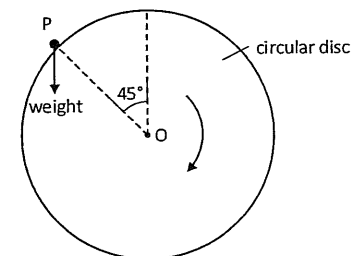


Fig. 1.2

- c. The angular velocity of the disc is increased gradually. The maximum value of the contact force between the disc and plasticine is 0.23 N.
 i. Explain why the plasticine is most likely to first lose contact with the disc at the lowest point of the revolution. [3]
 ii. Hence, determine the angular velocity when the plasticine first loses contact with the disc. [2]

[Total: 9]

2. [CJC/H2/Prelims 2018/P3/1]

A construction worker on the roof of a hemispherical dome releases a wrench at the highest point A with negligible speed as shown in Fig. 2.1. The radius R of the dome is 30.0 m. The surface of the dome's roof is smooth. At a certain point B, the wrench just loses contact with the surface of the dome and falls with a projectile motion through the air, and finally hits the ground at point C.

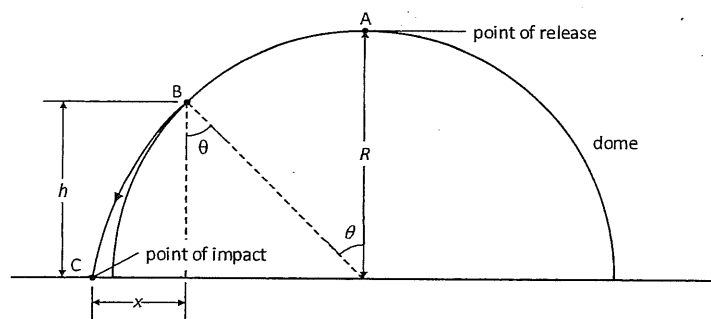


Fig. 2.1

- a. i. Explain why the centripetal acceleration of the wrench increases as it slides from A to B. [2]
 ii. State the magnitude of the normal contact force on the wrench at point B. [1]
 b. By considering the forces contributing to the centripetal force on the wrench at point B, show that the vertical distance between B and C, h is

$$h = \frac{v^2}{g}$$

where v is the linear velocity of the wrench at point B and g is the acceleration due to gravity. [3]

- c. i. At the point of losing contact with the surface of the dome, h is related to R by

$$h = \frac{2R}{3}$$

Use this relation and the relation in b. to calculate for the wrench at the point of losing,

1. its speed and [2]
 2. its direction of motion with respect to the horizontal. [2]
 ii. Using the equations of motion, determine the time taken for the wrench to fall from point B to C. Air resistance is assumed to be negligible. [3]
 iii. Hence determine x , the horizontal distance between points B and C. [2]

[Total: 13]

3. [DHS/H2/Prelims 2018/P3/4 Modified]

Explain,

- a. i. what is meant by a radian, [2]
 ii. why one complete revolution is equivalent to an angular displacement of 2π rad. [1]
 b. A stone of weight 3.0 N is fixed, using glue, to one end P of a rigid rod CP, as shown in Fig. 3.1.

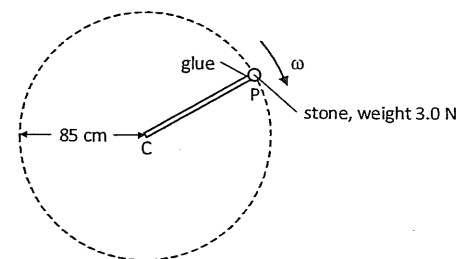


Fig. 3.1

The rod is rotated about end C so that the stone moves in a vertical circle of radius 85 cm.

The angular speed ω of the rod and stone is gradually increased from zero until the glue snaps. The glue fixing the stone snaps when the tension in it is 18.0 N.

For the position of the stone at which the glue snaps,

- i. mark with the letter S, the position of the stone on the dotted circle of Fig. 3.1 and [1]
 ii. calculate the angular speed ω of the stone. [3]
 c. The same stone is now fixed on a string and made to travel along a horizontal circular path, as shown in Fig. 3.2.

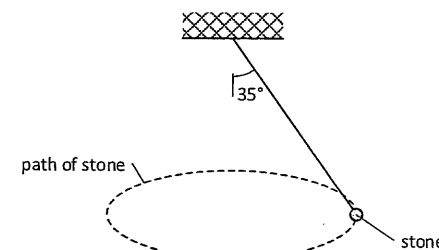


Fig. 3.2

The string makes an angle of 35° to the vertical.

Determine

- i. the tension T in the string, and [1]
 ii. the resultant force acting on the stone in the position shown. [2]

[Total: 10]

4. [JJC/H2/Prelims 2018/P3/3]

- a. An elastic light cord has an unextended length of 13.0 cm. One end of the cord is attached to a fixed point C. A small mass of 0.50 kg is hung from the free end of the cord. The cord extends to a length of 14.8 cm as shown in Fig. 4.1.

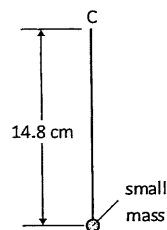


Fig. 4.1

- i. Determine the force constant k of the light cord. [1]
- ii. The mass is now made to move at constant angular speed ω in a vertical plane about point C. When the cord is vertical and above C, its length is the unextended length of 13.0 cm, as shown in Fig. 4.2.

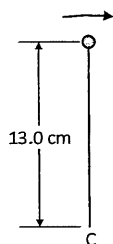


Fig. 4.2

Show that the angular speed ω of the mass is 8.7 rad s^{-1} . [2]

- iii. The mass moves so that the cord is vertically below C, as shown in Fig. 4.3.

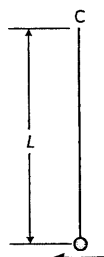


Fig. 4.3

Calculate the length L of the cord, assuming it obeys Hooke's law. [2]

- b. A mass m of 5.00 kg rests on the equator of the Earth as shown in Fig. 4.4. The Earth, assumed to be a perfect sphere, rotates about its axis with a period of 1.00 day. The surface of the Earth exerts a normal contact force N on the mass. The radius R of the Earth is $6.4 \times 10^3 \text{ km}$.

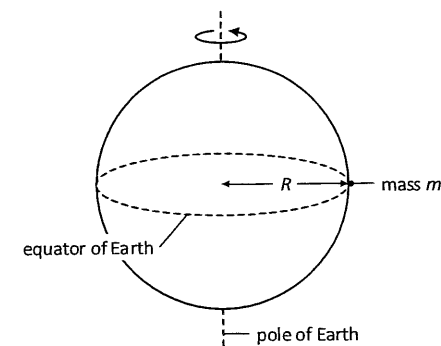


Fig. 4.4

- i. Calculate, for mass m ,
 1. the centripetal force F_c , [2]
 2. the normal contact force N . [2]
- ii. Explain why the normal contact force N acting on the mass at the poles is different from the answer in b.i.2. [2]

[Total: 11]

5. [NYJC/H2/Prelims 2018/P3/3]

At Luna Amusement Park in Sydney, Australia, the Rotor and Big Dipper Rollercoaster are the most exciting rides.

- a. The popular Rotor ride consists of a large vertical cylinder, which spins about its axis so quickly that any person standing inside is held against the wall when the floor drops away. The passengers maintain horizontal circular motion at constant speed within the cylinder. The normal contact force N acting on each passenger is related to the frictional force f on the passenger by a constant μ as given by the expression $f = \mu N$.

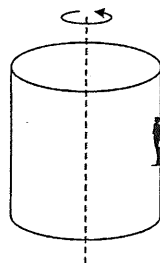


Fig. 5.1

- i. Draw and label, on Fig. 5.1, the forces acting on the passenger. [1]
 ii. Show that the linear velocity v at which the cylinder of radius R must rotate is independent of the mass of the passenger, m . [3]
- b. The cart on the exhilarating Big Dipper Rollercoaster moves with negligible friction along the track as shown in Fig. 5.2. The cart travels with an initial speed v_0 of 25 m s^{-1} at the top of one hill, of height h , before reaching the top of a second hill, which forms a circular arc of radius 95 m .

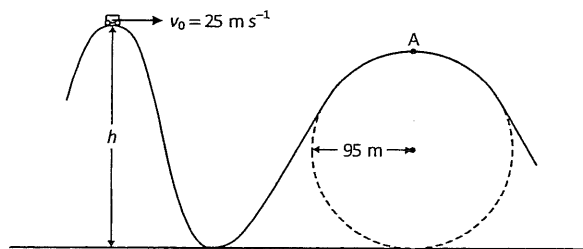


Fig. 5.2

- i. Calculate the maximum speed at which the cart can travel without leaving the track at the top of the second hill at point A. [2]
 ii. Hence, determine the maximum height h , of the first hill for the condition in b.i. to hold. [2]

[Total: 8]

6. [PJC/H2/Prelims 2018/P2/2]

- a. Explain how a body moving at constant speed can experience an acceleration. [2]
 b. A ball at rest at O is slightly displaced so that it slides down a frictionless track towards the loop-the-loop of radius R as shown in Fig. 6.1.

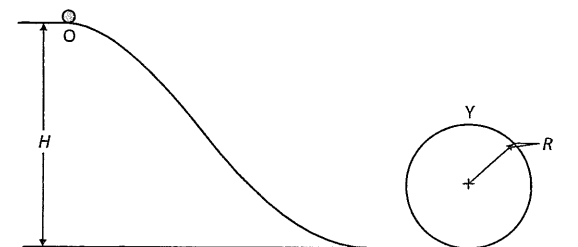


Fig. 6.1

- i. On Fig. 2.2, draw and label the forces acting on the ball when it is at the highest point Y of the loop-the-loop. [2]



Fig. 6.2

- ii. Show that the minimum speed of the ball at point Y so that it remains in contact with the track is $\sqrt{gR}$. [2]
 iii. Hence, determine an expression in terms of R for the minimum height H of point O with respect to the lowest point of the loop-the-loop. [2]

[Total: 8]

7. [RVHS/H2/Prelims 2018/P3/3 Modified]

A 0.050 kg mass is attached to one end of an elastic string of unstretched length 0.50 m . The other end of the string is attached to a fixed support. The force constant of the elastic string is 40 N m^{-1} . The mass is rotated steadily on a smooth table in a horizontal circle of radius 0.70 m as shown in Fig. 7.1.

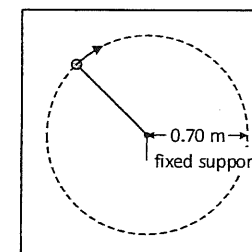


Fig. 7.1

Fig. 7.2 shows the side view of the string and mass.

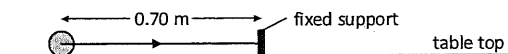


Fig. 7.2

- On Fig. 7.2, draw and identify the forces acting on the mass. [2]
- Calculate the linear speed of the mass. [2]
- The mass is now attached to a *non-elastic string* and rotated at a speed v in a vertical plane of radius r . Derive an expression, in terms of r , for the minimum speed at the highest point of the circular path at which the string *just* becomes slack. [2]

[Total: 6]

8. [VIC/H2/Prelims 2018/P3/8 Part]

- One end of a string is secured to the ceiling and a metal ball of mass 50 g is tied to its other end. The length of the string is 1.50 m. The ball is initially at rest in the vertical position. The ball is raised through a vertical height of 70 cm as shown in Fig. 8.1. When the ball is released, it describes a circular arc as it passes through the vertical position.

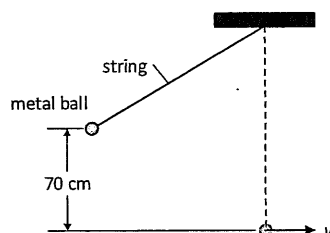


Fig. 8.1

- Ignoring the effects of air resistance, show that the speed v of the ball is 3.7 m s^{-1} as it passes through the vertical position. [1]
- Draw a labelled force diagram to show all the forces acting on the ball when the string is in the vertical position. [2]
- Calculate the centripetal force on the ball when it is in the vertical position. [2]
- Explain why the tension in the string when it is vertical is not equal to the weight of the ball. [2]
- Calculate the tension in the string when it is vertical. [2]

[Total: 9]

9. [YJC/H2/Prelims 2018/P3/2]

Fig. 9.1 shows two men, both with the same mass m , weighing themselves using identical weighing scales, at two different locations on the surface of the Earth. Both scales are in good working condition. Man P weighs himself at the North pole and has a reading of 589.2 N whereas man E weighs himself at the equator and has a reading of 587.4 N. The rotational acceleration of the Earth about an axis through its poles is given as a_c .

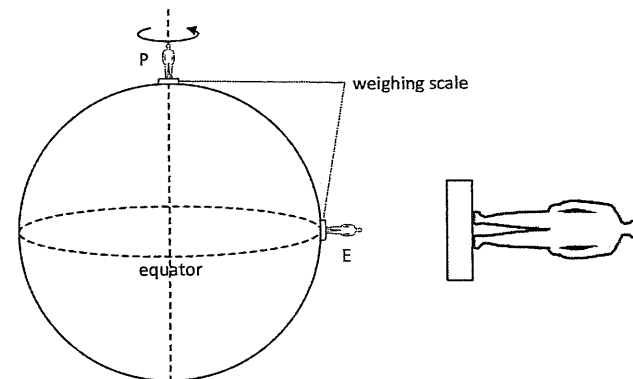


Fig. 9.1

Fig. 9.2

- Fig. 9.2 shows an enlarged diagram of man E and his weighing scale. On Fig. 9.2, sketch and label all the forces acting on man E. [2]
- Hence, or otherwise, explain why the scale readings at P and E are not the same. [3]
- Given that $m = 60 \text{ kg}$, deduce a_c . [2]

[Total: 9]

